

# Jin Woo Oh

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## SKILLS

**Languages & Frameworks:** Python, SQL, R, C++, TypeScript, Zero, Java, D3.js, Flask, Ollama, CUDA, RAPIDS

**Machine Learning & AI:** Claude, Modal, agent orchestration, Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-Learn, HuggingFace, LangChain, harness engineering, fine-tuning, hypothesis testing, regression, anomaly detection, deep learning

**Tools:** Git, Bash, Jupyter, PostgreSQL, Docker, Kubernetes, AWS, GCP, Dagster, Temporal, Apache, Power BI, BigQuery, MCP

## EXPERIENCE

**Avnir** **Nov 2025 – Present**  
*Software Engineer* *Atlanta, GA*

- Developed GeoPin, a location intelligence feature with self-hosted Nominatim on Kubernetes that recommends nearby contacts by goal-skill fit and discovers low-resistant warm introduction path via Apache AGE graph; exposed via MCP for in-house AI
- Built a LLM (Claude) contact extraction pipeline with parallelization across large uploaded files by decomposing Temporal activity into concurrent batches, reducing ingestion time from 40 minutes to 4 minutes
- Created a relationship-strength scoring SOM model end-to-end with 30 behavioral features tuned via grid search, stratified k-fold CV, and teacher-student score distillation; optimized pipeline reruns from 75 minutes to 5 seconds with per-user caching
- Automated the Customer Success reporting pipeline by replacing manual Google Sheet with 20-dataset/30-chart Apache Superset dashboard computing KPIs (DAU/WAU, engagement-depth scores, etc), deployed across 3 environments

**ICC Lab, Florida State University** **May 2025 – Nov 2025**  
*Researcher* *Tallahassee, FL (Remote)*

- Automated 1,000+ hours of manual labeling by building a gaze-tracking pipeline with Tobii eye-tracking data and OpenCV to overlay real-time gaze positions and regions of interest on screen recordings to streamline behavioral analysis
- Achieved 95% accuracy in detecting UI state change via robust template matching and HSV color masking

**FlightAware** **June 2024 – Aug 2024**  
*Software Engineer Intern* *Houston, TX (Remote)*

- Reduced latency of deep learning web application by 35% by developing scalable API (Kubernetes, Streamlit & FastAPI) to provide secure internal access to the web service across multiple teams and earning positive client feedback
- Enhanced robustness and improved system performance of the Foresight Labs website by 20% via streamlining REST API calls tested on Postman, reconfiguring dependencies on Redis and Triton servers
- Performed A/B test of GPT-4o vs Mistral 7B RAG chatbots; recommended GPT-4o for 21x speed & 40% higher success

**Samsung Electronics America** **Aug 2019 – Apr 2024**  
*Software Quality Engineer III* *Berkeley Heights, NJ*

- Spearheaded the development of a GPU-accelerated technical classifier model to identify customer complaints citing critical issues of flagship models with over 92% accuracy, automating 11+ hours of work per week
- Built and automated ETL pipeline (Python + SQL) processing 1M+ articles for downstream data classification and daily QA reporting of Android OS (9-14), improving data consistency and accelerating issue detection workflows by 25%
- Reduced operation cost of tech support live chat by 30% through a resource optimization simulation on Arena
- Presented clear, actionable business insights to stakeholders with user behavior & QA trend dashboards using Power BI

## PROJECTS

**Ahab** | *pytest, SQLite, GPU, Qwen/Gemma, llama.cpp, FastAPI, Tailscale, Slack SDK, launchd, API, CUSUM, TSMOM, XGBoost*

- Autonomous nightly equity-analysis engine applying flow signals, self-hosted LLM interpretation, and Slack alert

**RayTracing** | *C++, CUDA, linear algebra, computer graphics, parallel programming, GPU acceleration, multi-threading, HPC*

- Implemented CUDA ray tracer achieving 35x GPU speedup over CPU via parallel programming and HPC scaling

**Technical Issue Classifier** | *PyTorch, TensorFlow, Scikit-Learn, HuggingFace, CUDA, BeautifulSoup, Docker, Kubernetes, SQL*

- Optimized model by grid searching and threshold-tuning the ROC curve, minimizing costly false negatives to 3%

## EDUCATION

**Georgia Institute of Technology** **Expected Dec 2026**  
*Master of Science (M.S.) in Analytics* *Atlanta, GA (Online)*

**Seoul National University** **Feb 2018**  
*Master of Science (M.S.) in Mechanical Engineering* *Seoul, Korea*

- 2 patents, 2 journal publications, 8 conference presentations; Academic Scholarship; 3.75/4.0 GPA, *Summa cum laude*

**University of Illinois at Urbana-Champaign** **May 2015**  
*Bachelor of Science (B.S.) in Mechanical Engineering* *Champaign, IL*

- 1 journal publication; VP, Co-founder, mentor and tutor at Korean Mechanical Engineer Network; Academic Scholarship